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CIRCUITI E SISTEMI ELETTRONICI

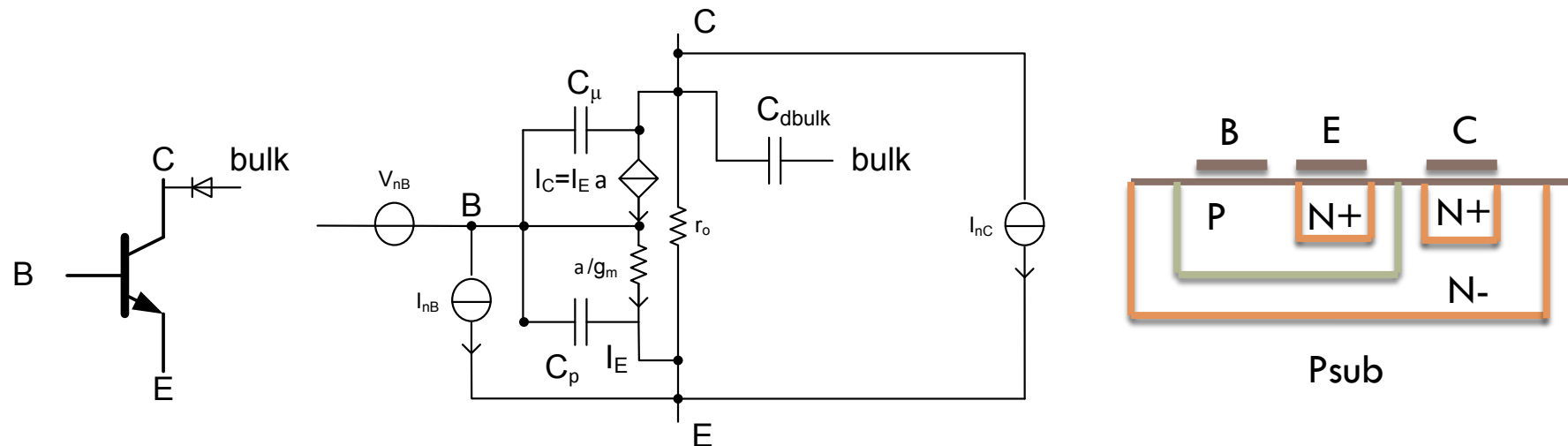
Stefano Sagginì
2 lezione Tecnologie analogiche

Different Type of MOS available in the technology

- Core MOS (minimum)
 - ▣ Small minimum dimension L or W
 - ▣ small maximum V_{gs} , V_{gd} and V_d
 - ▣ Can be version Low V_{TH} or Zero V_{TH}
- I/O MOS (in general defined By the V_{dd})
 - ▣ Higher minimum dimension L or W
 - ▣ Higher maximum V_{gs} , V_{gd} and V_d
- High Voltage LDMOS
 - ▣ Fixed L dimension
 - ▣ High V_{ds} voltage



BJT



Small signal

$$g_m = \frac{I_C}{V_T}$$

$$r_o = \frac{V_E}{I_C}$$

$$\alpha = \frac{\beta}{1 + \beta}$$

Approfondimento

Noise parameters

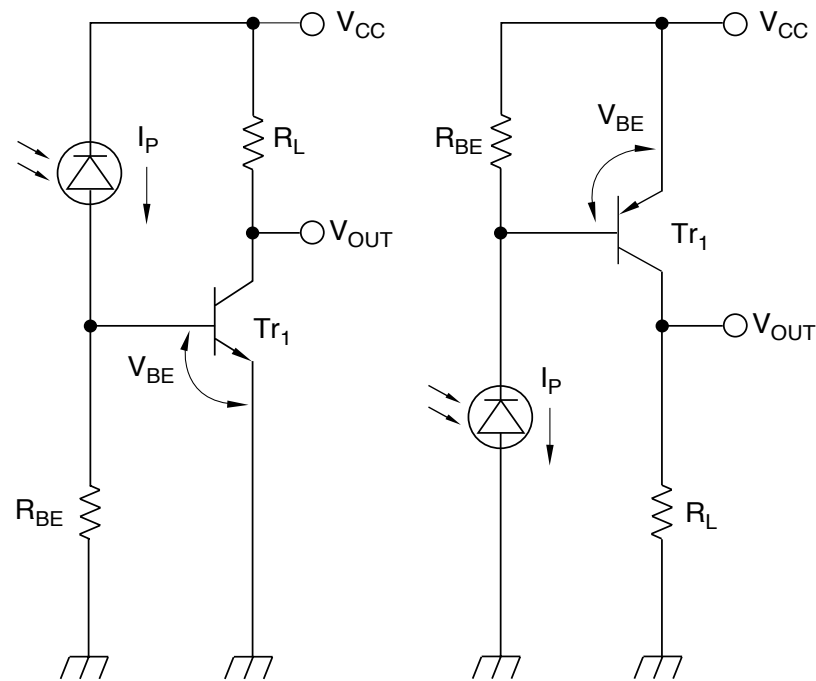
$$I_{nC} = 2qI_C$$

$$I_{nB} = 2qI_B$$

$$V_{nB} = 4kTR_b$$

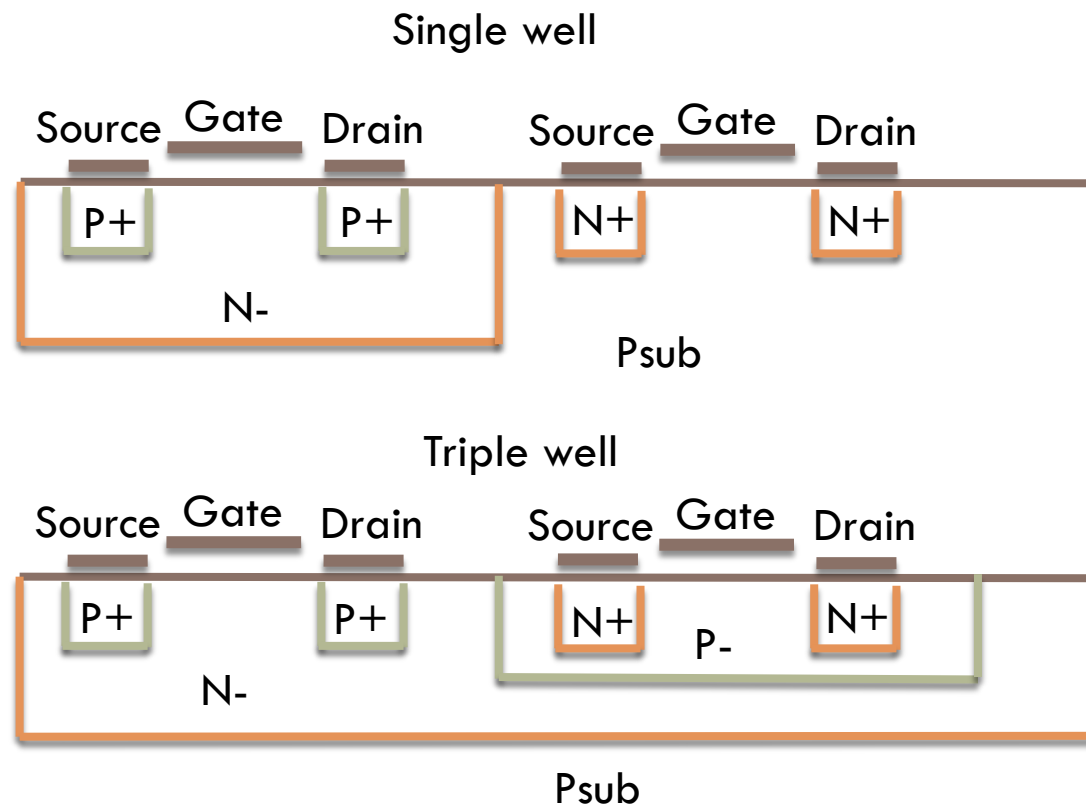
Our design first example

- Example of RX circuit with bipolar transistor



MOS model and description

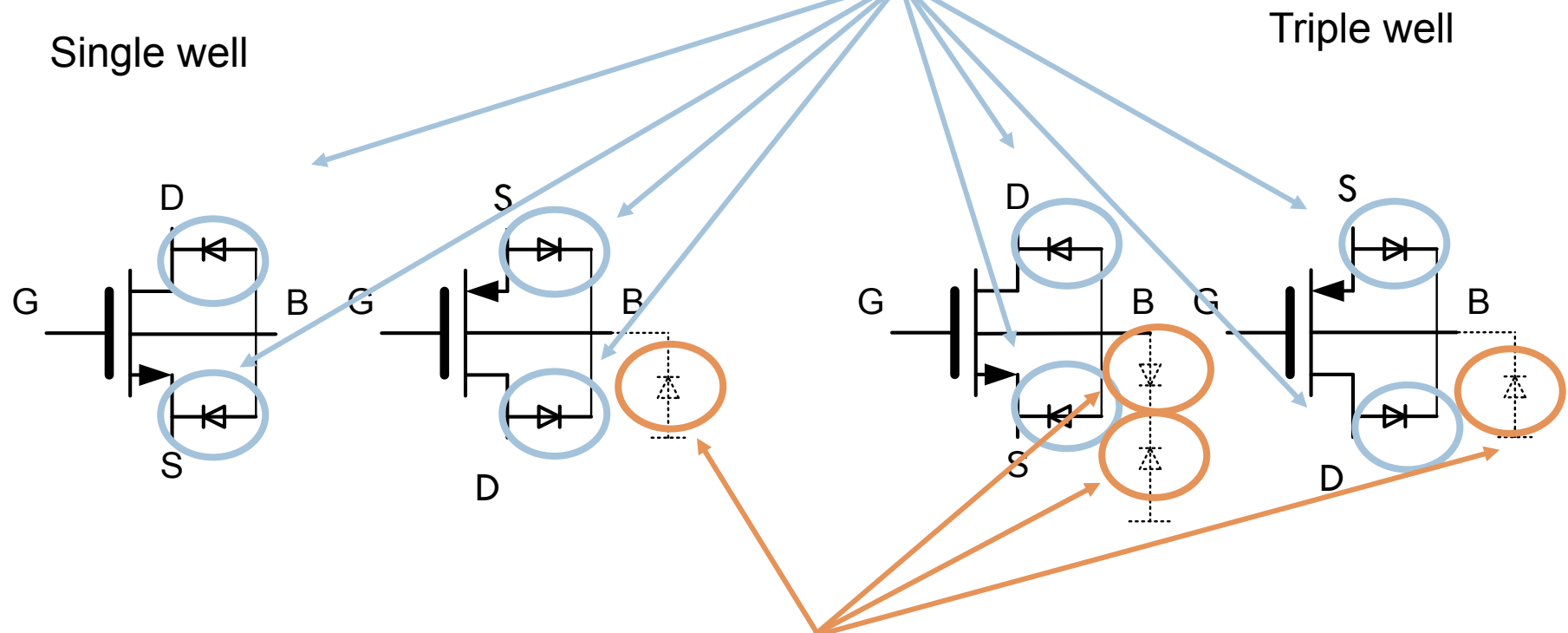
- Different types of MOS available from the technology



MOS SOA consideration

□ Voltage

Low voltage Diodes



These Diodes in many can have high BD voltage

MOS Description

□ MOS static model SPICE LEVEL 1

TRIODE

$$I_D = \mu_o C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) - \left(\frac{V_{DS}}{2} \right) \right] V_{DS} \quad (V_{GS} - V_{TH}) \geq 0 \quad (V_{GD} - V_{TH}) \geq 0$$

SATURATION

$$I_D = \mu_o C_{ox} \frac{W}{2L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS}) \quad (V_{GS} - V_{TH}) \geq 0 \quad (V_{GD} - V_{TH}) < 0$$

OFF STATE

$$I_D \cong 0 \quad (V_{GS} - V_{TH}) < 0 \quad (V_{GD} - V_{TH}) < 0$$

$$V_{TH} = V_{THO} + \gamma \left(\sqrt{2|\Phi_F| + V_{SB}} - \sqrt{2|\Phi_F|} \right)$$

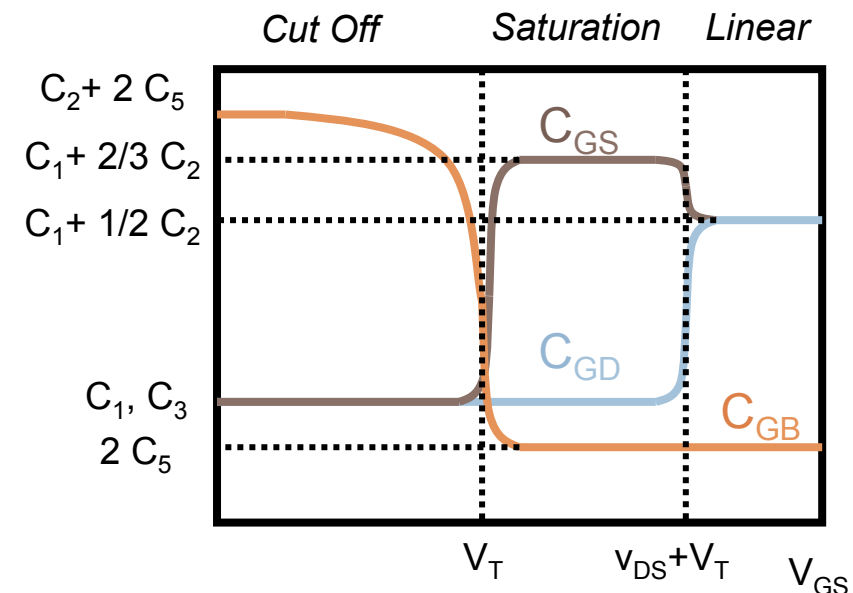
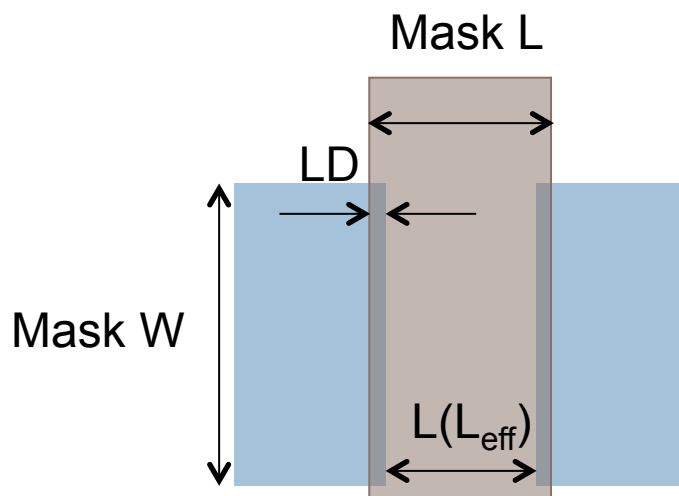
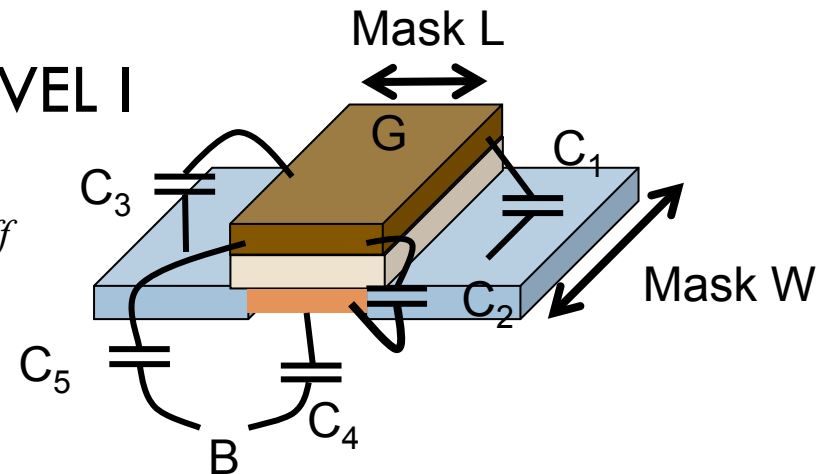
MOS Capacitors (AC)

□ MOS dynamic model SPICE LEVEL 1

$$C_1 = C_3 \cong (LD)C_{ox}W_{eff} = (CGXO)W_{eff}$$

$$C_2 = (L - 2LD)W_{eff}C_{ox}$$

$$2C_5 = CGB0(L_{eff})$$



Short channel effects SCE

- Output resistance decreases due to channel length modulation.
- V_T becomes more dependent on L_{eff} and V_D .
- I_{off} and subthreshold slope increase; susceptibility to punch-through increases.
- Vertical and lateral fields increase.
- Mobility degrades; velocity saturates; hot-carrier effects become more important.
- Overlap and fringe capacitances become a larger fraction of gate capacitance.
- Extrinsic resistance becomes a larger fraction of source-to-drain resistance.
- Flicker noise and mismatch increase

Saturation velocity

- When the E field reach a critical value the carrier reach the saturation velocity

$$\tau \approx \frac{L_{eff}}{v_{sat}}$$

- Hence the current became


$$I_D \approx \frac{W_{eff} L_{eff} \bar{Q}}{\tau} \approx W_{eff} C_{ox} (V_{GS} - V_T) v_{sat}$$

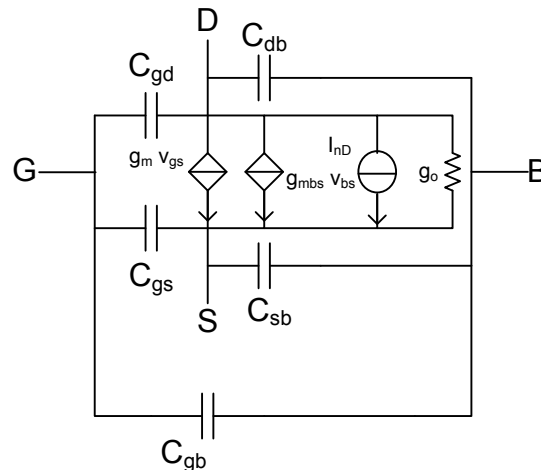
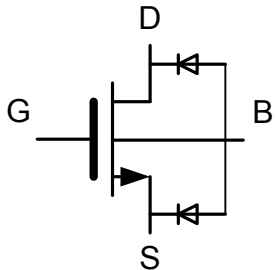
Average inversion charge for unit area

Small signal model in Saturation

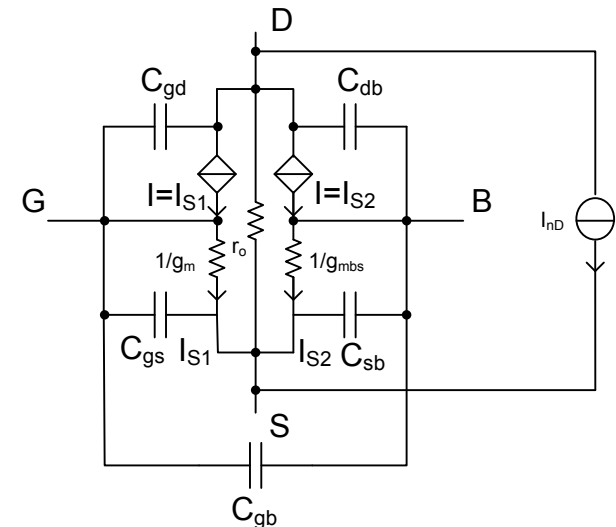
$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \sqrt{(2K'W/L)|I_D|(1+\lambda V_{DS})} = \sqrt{(2K'W/L)|I_D|}$$

$$g_{mbs} = -\frac{\partial I_D}{\partial V_{SB}} = -\frac{\partial I_D}{\partial V_T} \frac{\partial V_T}{\partial V_{SB}} = g_m \frac{\gamma}{2\sqrt{2|\Phi_F| + |V_{SB}|}} = \eta g_m$$

$$g_{ds} = g_o = \frac{\partial I_D}{\partial V_{SD}} = \frac{\lambda I_D}{1 + \lambda V_{DS}} \cong \lambda I_D \quad (\lambda \downarrow L \uparrow)$$



L'impedenza di ingresso
del dispositivo è
rigorosamente un circuito
aperto in Continua!!!!!!!!!!!!!!



Under threshold MOS current

$$i_D \cong \frac{W}{L} I_{DO} e^{\frac{v_{GS}}{n}}$$

$$1 < n < 3$$

$$I_{DO}$$

Process parameters

Temperature dependence

Temperature

$$V_{TH}(T) = V_{TH}(T_o) + TCV(T - T_o)$$

$$\ln(K(T)) = \ln(K(T_o)) + BEX(\ln(T) - \ln(T_o))$$

$$TCV \cong -1.5 \text{ [mV/K]}$$

$$BEX \cong -1.76$$

In saturation the temperature effect depends on V_{ov} .
For low overdrive voltage the current increase with the temperature with high overdrive is the contrary

Noise generator

Drain current noise generator

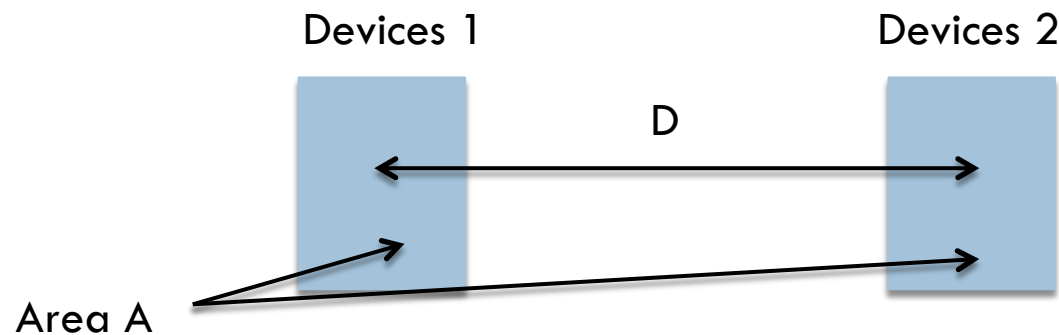
$$i_n^2 = \left[\frac{8kg_m(1+\eta)}{3} + \frac{KF I_D}{2f C_{OX} L^2} \right] \Delta f \quad (\text{A}^2)$$

Reported at the input

$$e_n^2 = \frac{i_n^2}{g_m^2} = \left[\frac{8k(1+\eta)}{3 g_m} + \frac{KF I_D}{2f C_{OX} WLK'} \right] \Delta f \quad (\text{V}^2)$$

Matching between two devices

- Considering two equal devices of area A on the same wafer characterized by the Parameter P (example the resistance, capacitor or the threshold voltage ecc..)



- P_1 and P_2 are two random variables
- The mean value of P on the wafer depend on process variation
- The difference $\Delta P = P_1 - P_2$ is Gaussian distributed with a variance

$$\sigma^2(\Delta P) = \frac{A_P^2}{A} + S_P^2 D^2$$

Experimentally determined

Matching between two devices

- In the technology manual is reported
 - ▣ Absolute variation between parameters

$$\Delta P = P_1 - P_2$$

- Used for V_T of MOS

- ▣ Relative variation

$$\frac{\Delta P}{P} = \frac{200(P_1 - P_2)}{P_1 + P_2}$$

- Used for K of MOS, Resistance and capacitors

- Only the coefficient related to the Area A

Matching between two MOS

$$\Delta P = P_1 - P_2$$

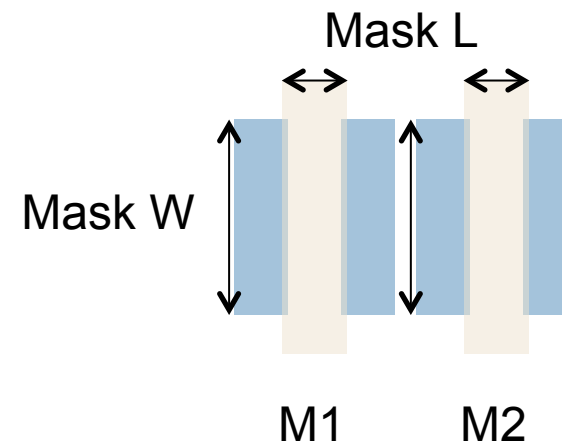
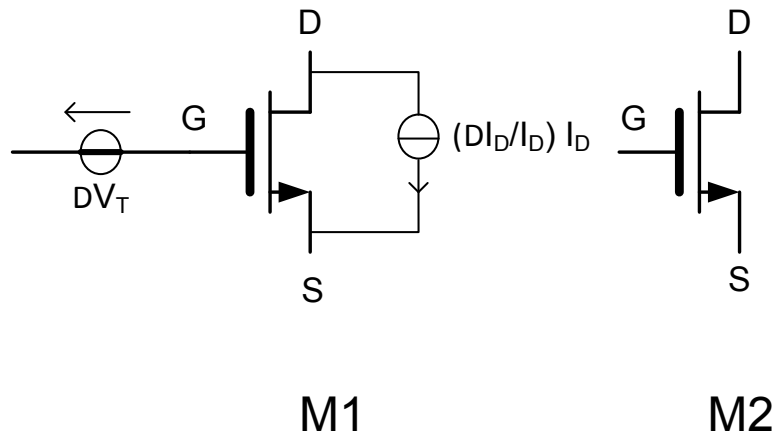
Used for V_T

$$\frac{\Delta P}{P} = \frac{200(P_1 - P_2)}{P_1 + P_2}$$

Used for K

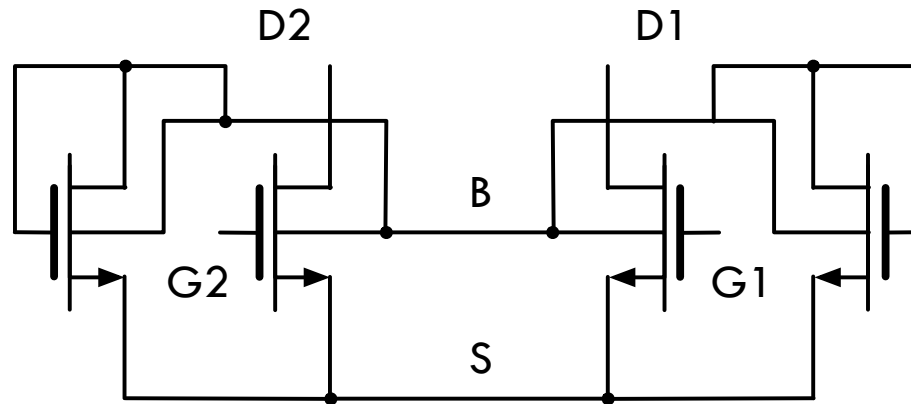
$$\sigma(\Delta P) = \frac{\alpha}{\sqrt{WL}} \quad \sigma\left(\frac{\Delta P}{P}\right) = \frac{\beta}{\sqrt{WL}}$$

Standard deviation

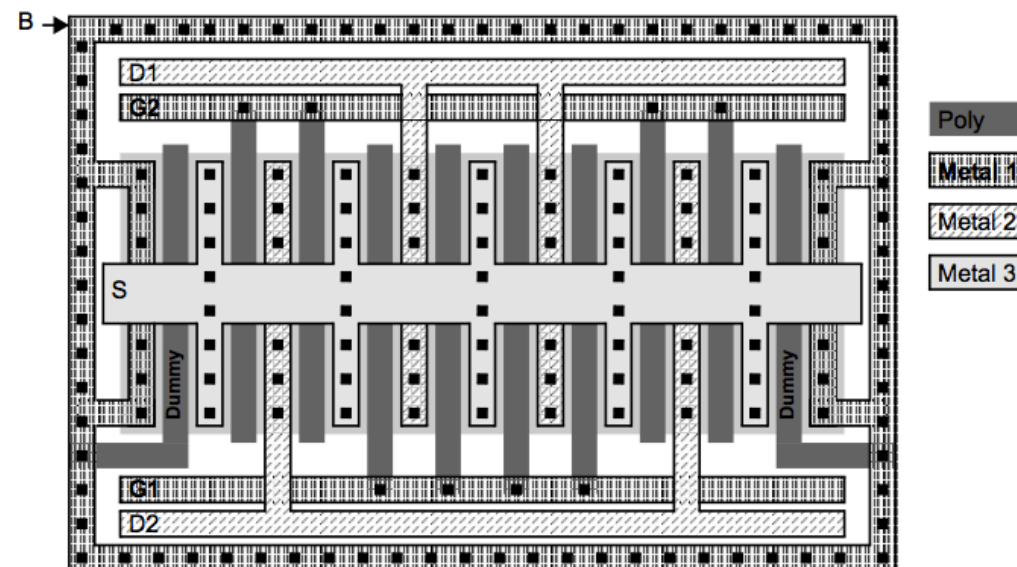


- Centroid

Schematiko

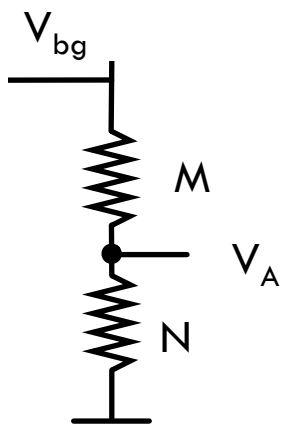


Layout



Examples for voltage dividers

□ Voltage references



$$V_A = \frac{M}{N + M} V_{bg}$$

Approfondimento

Calcolando la varianza della tensione otteniamo

$$\frac{\sigma V_A}{V_A} = \sqrt{\frac{M}{(N + M)N}} \frac{\sigma R}{R} \frac{1}{\sqrt{2}}$$

Scala con l'area